PERMANENCE OF WEAK ETALENESS FOR MAPS BETWEEN WEAKLY ETALE ALGEBRAS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a weakly etale A-algebra B the map B is flat over B tensor_A B, and flatness transfers along surjective ring maps with the module flat over the source. invoked at proof lines 30411-30414. Provides both the flatness of B -> C and the transfer to the quotient.

Source reference. Original lemma 092C: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30202–30202. Cited at proof lines 30411-30414 for both the flatness of B -> C and the transfer along the surjection; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a ring. Let $B \to C$ be an A-algebra map of weakly étale A-algebras. Then $B \to C$ is weakly étale.*

Proof. The ring map $B \to C$ is flat by Lemma 092C. The ring map $C \otimes_A C \to C \otimes_B C$ is surjective, hence an epimorphism. Thus Lemma 092C implies, that since C is flat over $C \otimes_A C$ also C is flat over $C \otimes_B C$. □

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